SRI RAMACHANDRA
INSTITUTE OF HIGHER EDUCATION AND RESEARCH
(Category - I Deemed to be University) Porur, Chennai
SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

QueryMind — Natural Language to SQL Converter

CSC23DLU05 Natural Language Processing

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report on **"NATURAL LANGUAGE TO SQL CONVERTER"** is the bonafide record of work done by **"KOMALAVALLI S - E5223009"** who carried out the internship work under my supervision.

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Signature of the Internal Examiner **Signature of the External Examiner**

Evaluation Date: 24-03-26

ABSTRACT

The QueryMind project aims to convert natural language queries into SQL queries using machine learning techniques. In many organizations, databases are used to store large volumes of data, but accessing this data requires knowledge of Structured Query Language (SQL). Non-technical users often find it difficult to write SQL queries, which limits their ability to retrieve useful information. This project addresses this problem by allowing users to interact with databases using simple English questions.

The system uses Natural Language Processing (NLP) to understand the input query and convert it into a structured SQL statement. TF-IDF (Term Frequency–Inverse Document Frequency) is used for feature extraction, which transforms text into numerical vectors based on word importance. A Linear Support Vector Machine (Linear SVM) is used as the classification model to predict the correct SQL query from the input.

The model is trained using a dataset that contains pairs of natural language questions and their corresponding SQL queries. After training, the system can generate SQL queries for new inputs with reasonable accuracy. This makes the process of querying databases faster and easier.

The main goal of this project is to reduce the complexity of database interaction and make it accessible to all users, regardless of their technical background. This approach provides a simple, efficient, and scalable solution, with future scope for improvement using advanced deep learning techniques.

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1. INTRODUCTION

In today's data-driven world, databases play an important role in storing and managing large amounts of information across various domains such as healthcare, education, and business. To retrieve data from these databases, users must write queries using Structured Query Language (SQL). However, writing SQL queries requires technical knowledge, and many non-technical users face difficulties in understanding and using it effectively.

To overcome this challenge, Natural Language Processing (NLP) techniques can be used to allow users to interact with databases using simple English language. The main idea is to convert natural language queries into SQL queries automatically, making data access easier and more efficient.

In this project, we developed a system called QueryMind, which translates user questions written in plain English into corresponding SQL queries. The system uses TF-IDF (Term Frequency–Inverse Document Frequency) to convert textual data into numerical features, capturing the importance of words in the input. A Linear Support Vector Machine (Linear SVM) algorithm is used to classify and predict the most appropriate SQL query based on the input question.

The model is trained using a dataset containing pairs of natural language questions and their corresponding SQL queries. Once trained, it can generate SQL queries for new inputs with good accuracy.

This project aims to simplify database interaction, reduce dependency on SQL knowledge, and provide a user-friendly solution for efficient data retrieval.

1.1 LINEAR SVM

Linear SVM is used in this project to classify natural language queries into their corresponding SQL queries efficiently and accurately. Since the input data is text-based, it is first converted into numerical form using TF-IDF, which creates high-dimensional feature vectors. Linear SVM is well-suited for handling such high-dimensional data and can effectively separate different query patterns using a linear decision boundary. The main purpose of using Linear SVM is to achieve good accuracy while keeping the model simple, fast, and computationally efficient. It helps in correctly predicting the structure of SQL queries from user input, making the system reliable for real-time usage. Compared to more complex models, Linear SVM requires less training time and performs well even with limited data, making it an ideal choice for this project.

2. OBJECTIVE

- To convert natural language into SQL queries
- To make database access easy for non-technical users
- To reduce dependency on SQL knowledge
- To improve accuracy in query generation
- To use TF-IDF for effective text processing
- To apply Linear SVM for fast and efficient classification

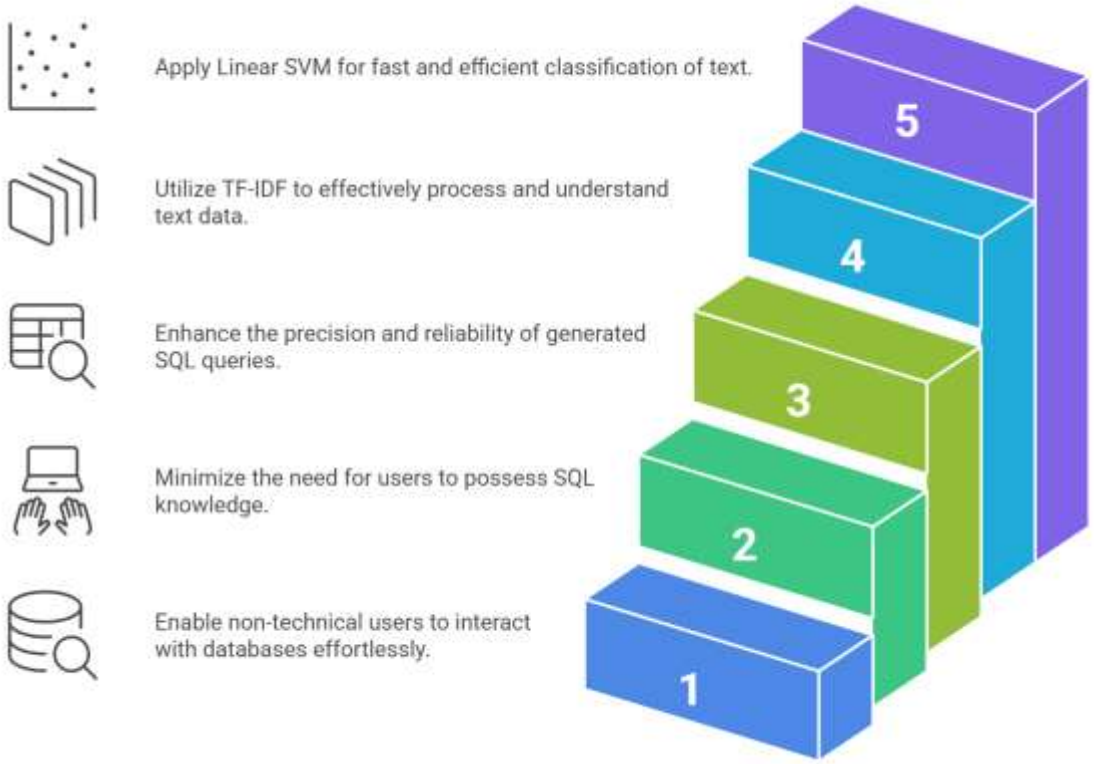

Fig 2.1 Objective

3. PROBLEM STATEMENT

- Users find it difficult to write SQL queries
- Requires technical knowledge of SQL syntax
- Non-technical users cannot easily access databases
- Data retrieval becomes time-consuming and complex
- High chance of errors in manual query writing
- Gap between users and database systems
- Need for a solution to convert natural language into SQL

5.DATASET DESCRIPTION

	DESCRIPTION
Dataset Name	WikiSQL Spider
Year	2017-2018
Dataset Source	Kaggle
Dataset Type	Text Dataset
Attribute	2 (Question, Sql)
Rows	88648
Usage	Training and testing NL2SQL model

6.METHODOLOGY

- **Data Collection** :Collect a dataset containing natural language questions and corresponding SQL queries.
- **Data Preprocessing** :Clean the text by converting to lowercase and removing unwanted symbols or noise.
- **Feature Extraction (TF-IDF)** :Convert the cleaned text into numerical vectors using TF-IDF to represent word importance.
- **Model Training (Linear SVM)** :Train the Linear SVM model using the TF-IDF features and labeled SQL queries.
- **Model Testing** :Test the model with new input questions to check prediction accuracy.
- **Prediction** :The trained model converts user input (English query) into the correct SQL query.

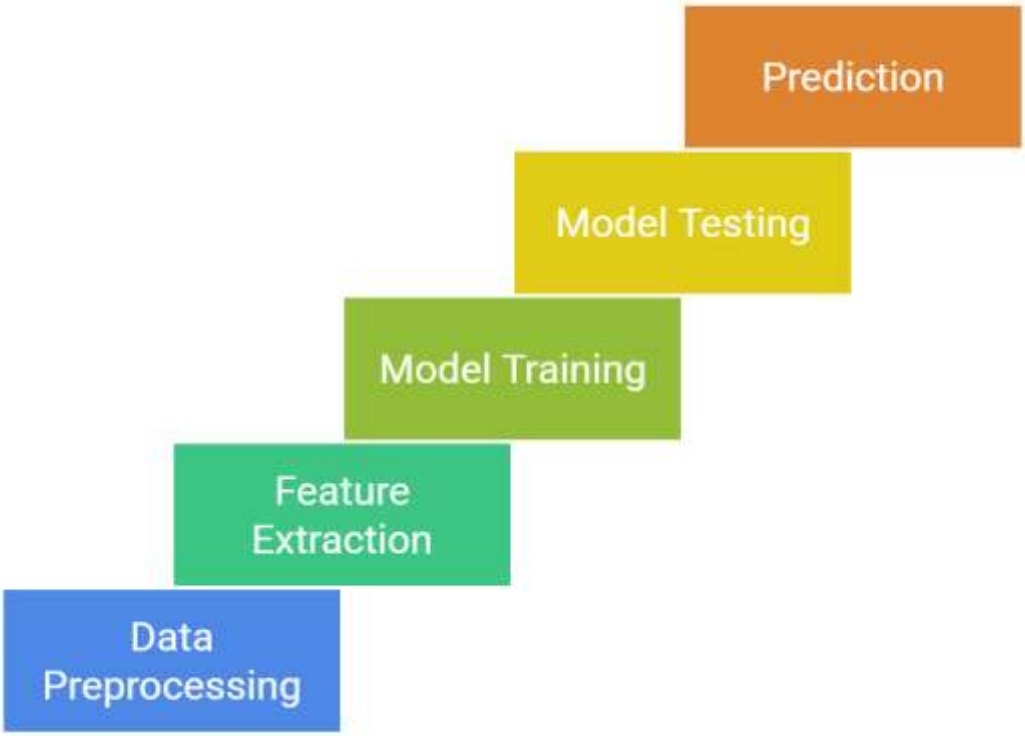

Fig 6.1 Methodology

8.RESULTS AND DISCUSSION

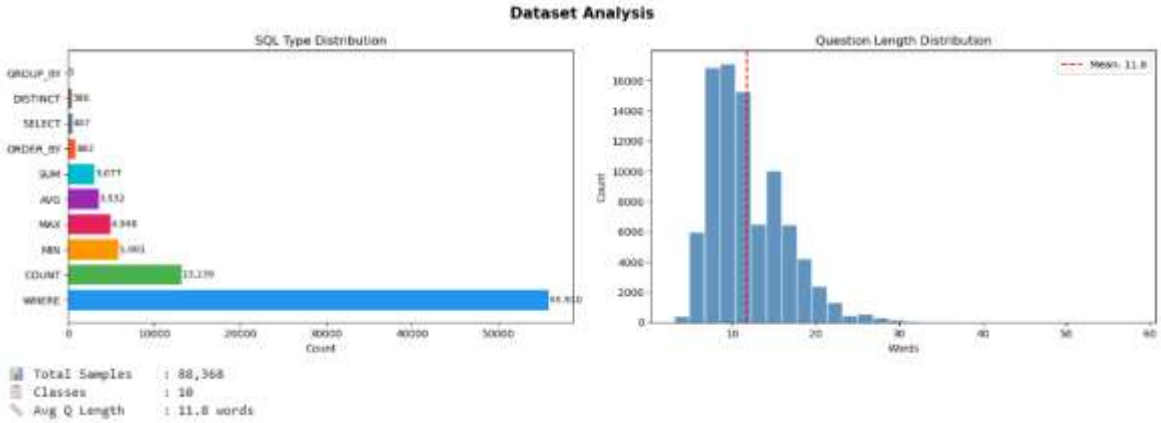

Fig 8.1 Dataset Analysis

```
🔍 Sample preprocessing:
================================================================
Original : Tell me what the notes are for South Australia
Cleaned  : tell me what the notes are for south australia
Label    : WHERE

Original : What is the current series where the new series began in June 2011?
Cleaned  : what is the current series where the new series began in june 2011
Label    : WHERE

Original : What is the format for South Australia?
Cleaned  : what is the format for south australia
Label    : WHERE

Original : Name the background colour for the Australian Capital Territory
Cleaned  : name the background colour for the australian capital territory
Label    : WHERE

Original : how many times is the fuel propulsion is cng?
Cleaned  : how many times is the fuel propulsion is cng
Label    : COUNT
```

Fig 8.2 Preprocessing

```
🚀 Training all models...
==================================================
⏳ Training Linear SVM...
  ✅ Linear SVM Accuracy: 87.46%
```

Fig 8.3 Accuracy

🏆 Best Model : Linear SVM
🎯 Accuracy : 87.46%

📋 Classification Report:

	precision	recall	f1-score	support
AVG	0.84	0.71	0.77	707
COUNT	0.79	0.78	0.79	2648
DISTINCT	0.76	0.62	0.69	77
GROUP_BY	0.00	0.00	0.00	1
MAX	0.87	0.73	0.79	990
MIN	0.89	0.71	0.79	1180
ORDER_BY	0.88	0.81	0.85	177
SELECT	0.80	0.59	0.68	97
SUM	0.64	0.40	0.50	615
WHERE	0.90	0.97	0.93	11182
accuracy			0.87	17674
macro avg	0.74	0.63	0.68	17674
weighted avg	0.87	0.87	0.87	17674

Fig 8.4 Classification Report

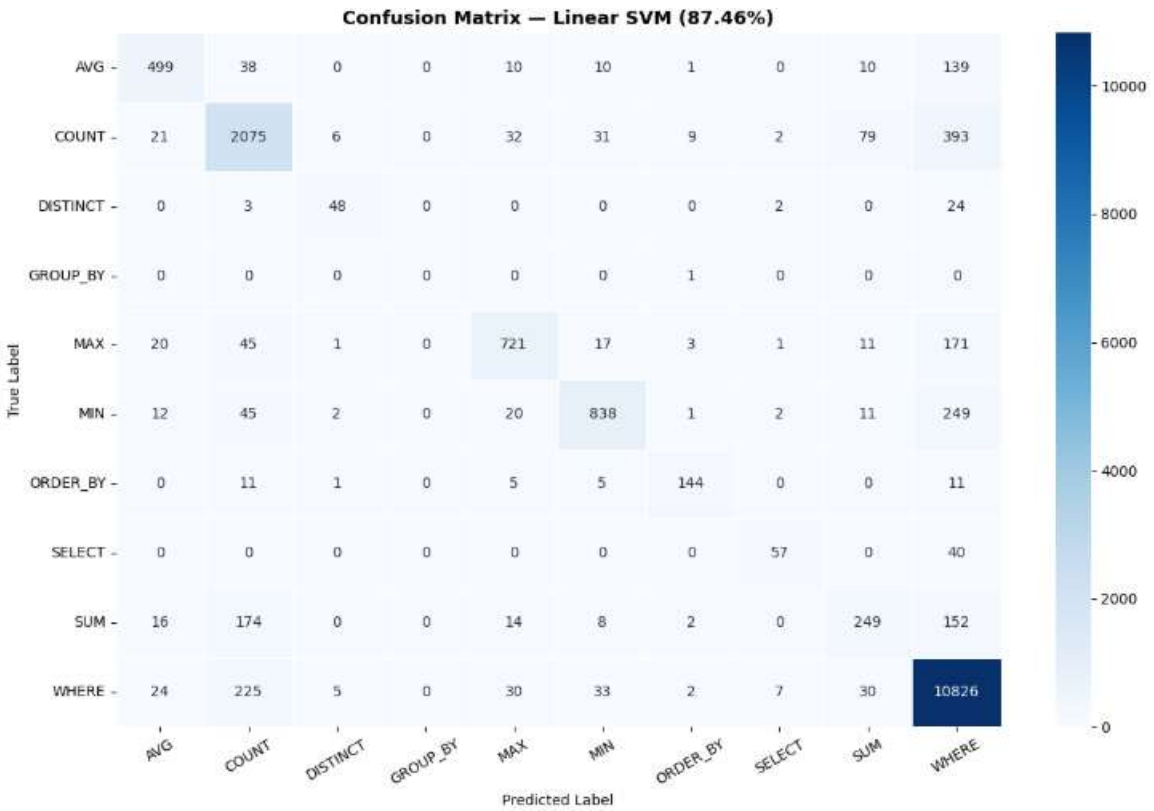

Fig 8.5 Confusion Matrix

```
[16]: nl2sql_demo("What is the maximum salary?")

      ==================================================================
        Question  : What is the maximum salary?
         SQL Type  : MAX
        SQL Query : SELECT MAX(column) FROM table_name
      ==================================================================
[16]: ('MAX', 'SELECT MAX(column) FROM table_name')

[17]: nl2sql_demo("Show all products ordered by price")

      ==================================================================
        Question  : Show all products ordered by price
         SQL Type  : ORDER_BY
        SQL Query : SELECT * FROM table_name ORDER BY column DESC
      ==================================================================
[17]: ('ORDER_BY', 'SELECT * FROM table_name ORDER BY column DESC')

[18]: nl2sql_demo("What is the average marks of students?")

      ==================================================================
        Question  : What is the average marks of students?
         SQL Type  : AVG
        SQL Query : SELECT AVG(column) FROM table_name
      ==================================================================
[18]: ('AVG', 'SELECT AVG(column) FROM table_name')
```

Fig 8.6 Sample Input

9.FUTURE WORK

- Improve model accuracy using deep learning techniques (LSTM, Transformers)
- Support complex SQL queries like JOIN, GROUP BY, and nested queries
- Add voice input for user queries
- Connect with real-time databases for live data access
- Develop a web or mobile application for easy usage
- Enhance dataset size for better training and performance
- Add multi-language support (Tamil, Hindi, etc.)
- Improve user interface for better user experience

10.CONCLUSION

In this project, we developed a system called QueryMind that converts natural language queries into SQL queries using machine learning techniques. The main goal was to help non-technical users interact with databases without needing to learn SQL. By using TF-IDF for feature extraction and Linear SVM for classification, the model is able to understand user input and generate appropriate SQL queries with good accuracy.The project demonstrates how Natural Language Processing (NLP) can be applied to solve real-world problems and simplify complex tasks like database querying. It reduces the effort required to write SQL queries manually and minimizes errors, making data retrieval faster and more efficient.Although the model performs well for simple queries, there is still scope for improvement in handling complex queries and improving accuracy. With further enhancements, this system can be extended to real-time applications such as business analytics tools and database management platforms.

Overall, this project provides a practical, user-friendly, and efficient solution for bridging the gap between human language and database systems, making data access easier for everyone.

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